Estimating Fundamental Matrix

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Topic: Uncalibrated Stereo, Module: Reconstruction II

First Principles of Computer Vision

Stereo Calibration Procedure

Find a set of corresponding features in left and right images (e.g. using SIFT or hand-picked)

Left image



Right image



Stereo Calibration Procedure

Find a set of corresponding features in left and right images (e.g. using SIFT or hand-picked)

Left image



Right image



- $(u_l^{(1)}, v_l^{(1)})$

⋮

- $(u_l^{(m)}, v_l^{(m)})$

- $(u_r^{(1)}, v_r^{(1)})$

⋮

- $(u_r^{(m)}, v_r^{(m)})$

Stereo Calibration Procedure

Step A: For each correspondence i, write out epipolar constraint.

$$\begin{bmatrix} u_l^{(i)} & v_l^{(i)} & 1 \end{bmatrix} \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{bmatrix} u_r^{(i)} \\ v_r^{(i)} \\ 1 \end{bmatrix} = 0$$

Known Unknown Known

Stereo Calibration Procedure

Step A: For each correspondence i, write out epipolar constraint.

$$\begin{bmatrix} u_l^{(i)} & v_l^{(i)} & 1 \end{bmatrix} \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{bmatrix} u_r^{(i)} \\ v_r^{(i)} \\ 1 \end{bmatrix} = 0$$

Known Unknown Known

Expand the matrix to get linear equation:

$$\left(f_{11}u_r^{(i)} + f_{12}v_r^{(i)} + f_{13}\right)u_l^{(i)} + \left(f_{21}u_r^{(i)} + f_{22}v_r^{(i)} + f_{23}\right)v_l^{(i)} + f_{31}u_r^{(i)} + f_{32}v_r^{(i)} + f_{33} = 0$$

Stereo Calibration Procedure

Step B: Rearrange terms to form a linear system.

$$\begin{bmatrix} u_l^{(1)}u_r^{(1)} & u_l^{(1)}v_r^{(1)} & u_l^{(1)} & v_l^{(1)}u_r^{(1)} & v_l^{(1)}v_r^{(1)} & v_l^{(1)} & u_r^{(1)} & v_r^{(1)} & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ u_l^{(i)}u_r^{(i)} & u_l^{(i)}v_r^{(i)} & u_l^{(i)} & v_l^{(i)}u_r^{(i)} & v_l^{(i)}v_r^{(i)} & v_l^{(i)} & u_l^{(i)} & u_r^{(i)} & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ u_l^{(m)}u_r^{(m)} & u_l^{(m)}v_r^{(m)} & u_l^{(m)} & v_l^{(m)}u_r^{(m)} & v_l^{(m)}v_r^{(m)} & v_l^{(m)} & u_l^{(m)} & u_r^{(m)} & 1 \end{bmatrix} \begin{bmatrix} f_{11} \\ f_{21} \\ f_{31} \\ f_{21} \\ f_{22} \\ f_{23} \\ f_{31} \\ f_{32} \\ f_{33} \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

A
(Known)

$\mathbf{f}$
(Unknown)

$$A\,\mathbf{f} = \mathbf{0}$$

The Tale of Missing Scale

Fundamental matrix acts on homogenous coordinates.

$$\begin{bmatrix} u_l & v_l & 1 \end{bmatrix} \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} = 0 = \begin{bmatrix} u_l & v_l & 1 \end{bmatrix} \begin{bmatrix} kf_{11} & kf_{12} & kf_{13} \\ kf_{21} & kf_{22} & kf_{23} \\ kf_{31} & kf_{32} & kf_{33} \end{bmatrix} \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix}$$

Fundamental Matrix F and kF describe the same epipolar geometry. That is, F is defined only up to a scale.

Set Fundamental Matrix to some arbitrary scale.

$$\|\mathbf{f}\|^2 = 1$$

Solving for F

Step C: Find least squares solution for fundamental matrix F.

We want $A\mathbf{f}$ as close to 0 as possible and $\|\mathbf{f}\|^2 = 1$:

$$\min_{\mathbf{f}} \|A\mathbf{f}\|^2 \quad \text{such that } \|\mathbf{f}\|^2 = 1$$

Constrained linear least squares problem

Like solving Projection Matrix during Camera Calibration.

Or, Homography Matrix for Image Stitching.

Rearrange solution $\mathbf{f}$ to form the fundamental matrix F.

Extracting Rotation and Translation

Step D: Compute essential matrix E from known left and right intrinsic camera matrices and fundamental matrix F.

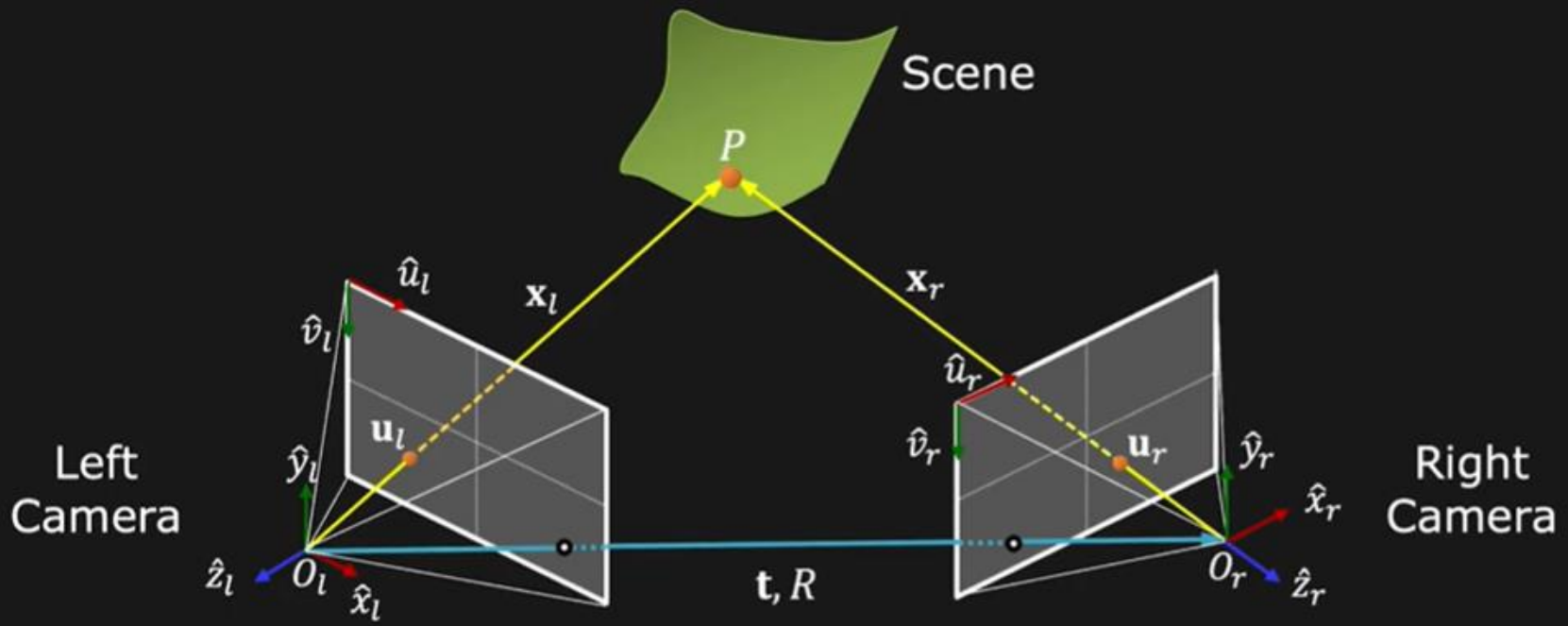
$$E = K_l^T F K_r$$

Step E: Extract R and $\mathbf{t}$ from E.

$$E = T_\times R$$

(Using Singular Value Decomposition)

Uncalibrated Stereo



1. Assume Camera Matrix K is known for each camera
2. Find a few Reliable Corresponding Points
3. Find Relative Camera Position $\mathbf{t}$ and Orientation R
4. Find Dense Correspondence
5. Compute Depth using Triangulation